

REGULATION OF PANCREATIC SECRETION

HEADINGS

- ▼ REGULATION OF PANCREATIC SECRETION
 - STAGES OF PANCREATIC SECRETION
 - ▼ 1. CEPHALIC PHASE
 - A. Unconditioned Reflex
 - B. Conditioned Reflex
 - 2. GASTRIC PHASE
 - ▼ 3. INTESTINAL PHASE
 - ▼ HORMONES STIMULATING PANCREATIC SECRETION
 - 1. Secretin
 - 2. Cholecystokinin (CCK / Cholecystokinin–Pancreozymin / CCK-PZ)
 - HORMONES INHIBITING PANCREATIC SECRETION
 - SCHEMATIC REGULATION SUMMARY

- Secretion of pancreatic juice is regulated by both **nervous and hormonal factors**.

STAGES OF PANCREATIC SECRETION

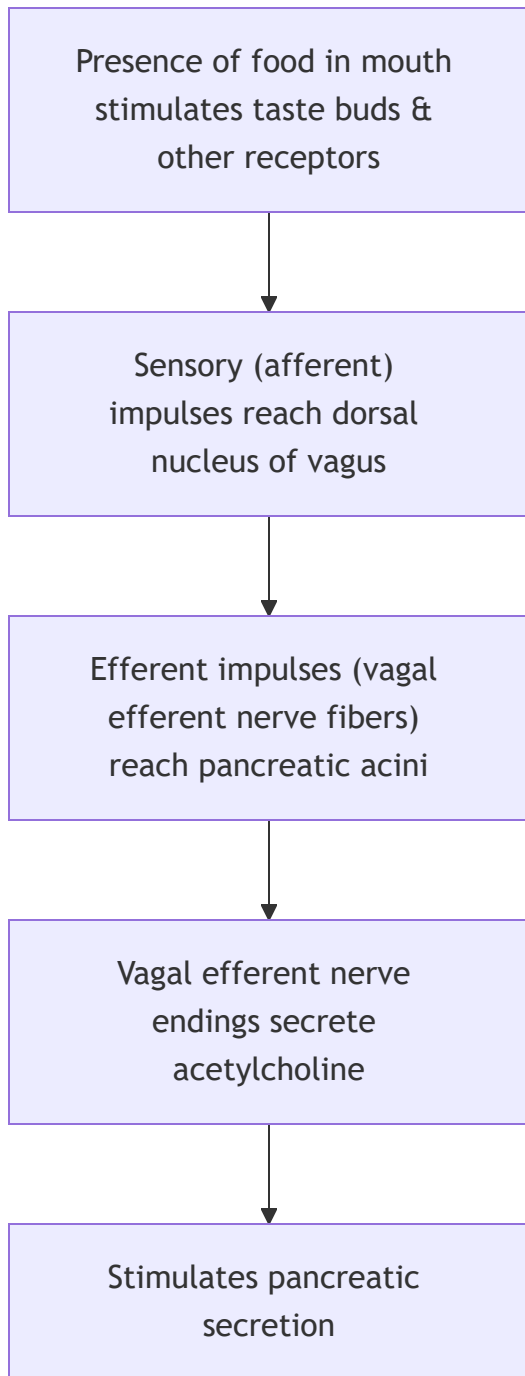
Occurs in **3 stages** (*corresponds with the 3 phases of gastric secretion*):

- a. Cephalic phase
- b. Gastric phase
- c. Intestinal phase

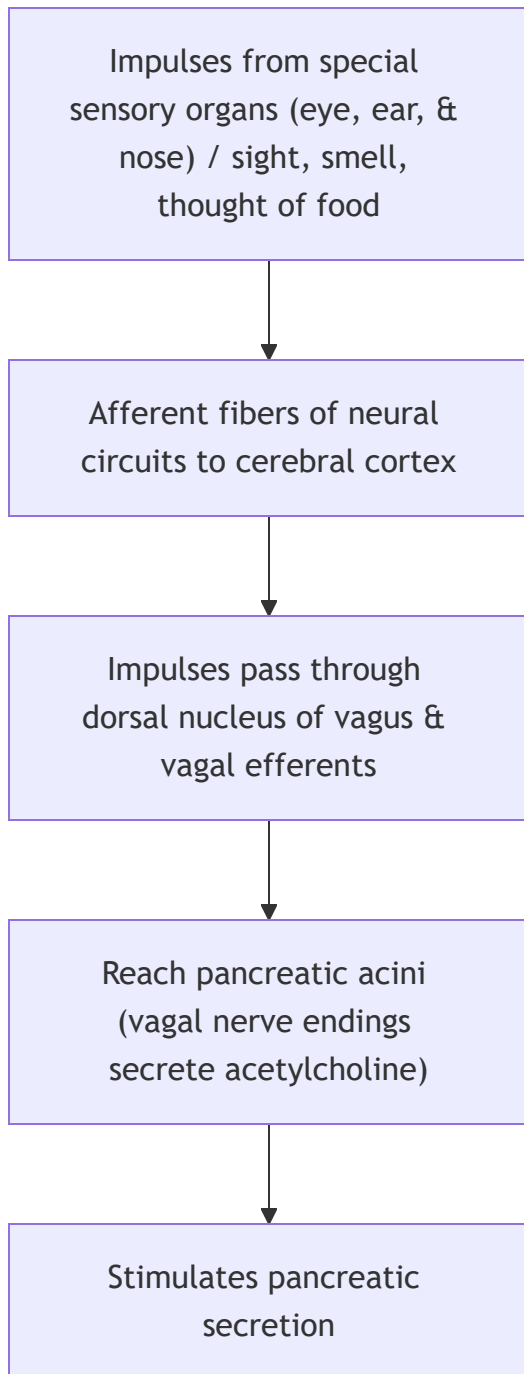
1. CEPHALIC PHASE

- Regulated by **nervous mechanism through reflex action**.
- **2 Types :—**
 - a. Unconditioned reflex
 - b. Conditioned reflex

A. Unconditioned Reflex

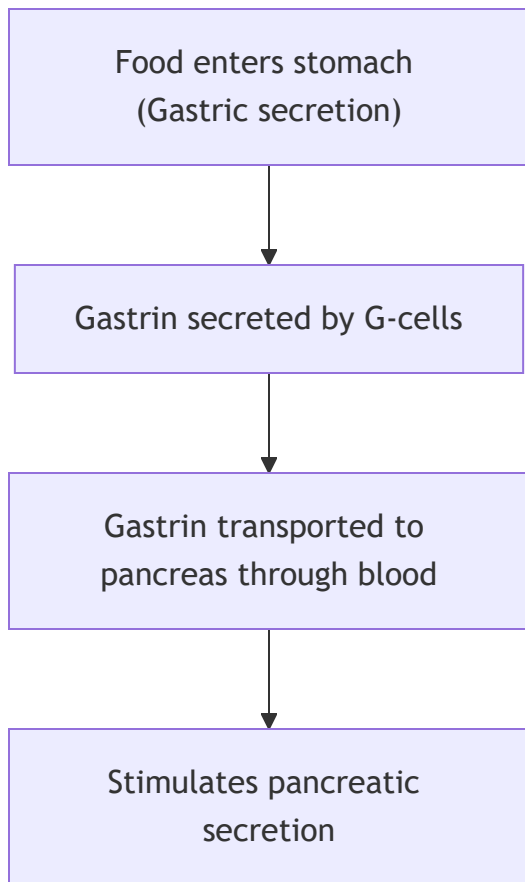


B. Conditioned Reflex



2. GASTRIC PHASE

- **Secretion of pancreatic juice** when food enters the stomach.
- This phase is under **hormonal control (Gastrin)**.



3. INTESTINAL PHASE

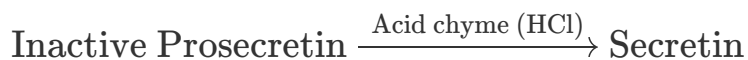
- **Secretion of pancreatic juice** when chyme enters the intestine.
- This phase is also under **hormonal control**:
- Some hormones **stimulate** pancreatic secretion.
- Some hormones **inhibit** pancreatic secretion.

HORMONES STIMULATING PANCREATIC SECRETION

- a. **Secretin**
- b. **Cholecystokinin (CCK / CCK-PZ)**

1. Secretin

- **Production** : Produced by **S-cells** of mucous membrane in duodenum & jejunum.



- **Actions of Secretin** :—
 - a. Stimulates secretion of **watery juice** (*rich in bicarbonate ions and high in volume*).
 - b. Increases pancreatic secretion by acting on **pancreatic ductules** via **cyclic AMP (cAMP)** as second messenger.

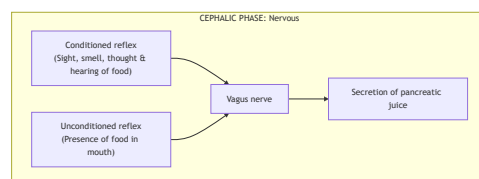
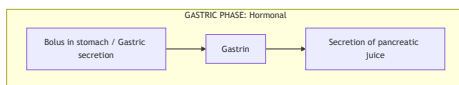
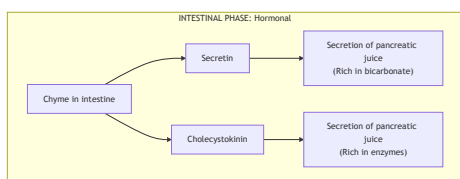
2. Cholecystokinin (CCK / Cholecystokinin–Pancreozymin / CCK-PZ)

- **Production** : Secreted by **I-cells** in duodenal & jejunal mucosa.
- **Stimulant for Release** : Chyme containing digestive products such as **fatty acids (FA)**, **peptides**, and **amino acids (AA)**.
- **Actions of Cholecystokinin** :—
 - a. Stimulates secretion of **pancreatic juice** (*rich in enzymes and very low in volume*).
 - b. Acts on **pancreatic acinar cells** via **Inositol Triphosphate (IP₃)** as second messenger.

HORMONES INHIBITING PANCREATIC SECRETION

- a. **Pancreatic Polypeptide (PP)** :— Secreted by PP cells in islets of Langerhans of pancreas.
- b. **Somatostatin** :— Secreted by D-cells in islets of Langerhans of pancreas.
- c. **Peptide YY** :— Secreted by intestinal mucosa.
- d. **Other Peptides** :— Such as ghrelin & leptin.

SCHEMATIC REGULATION SUMMARY



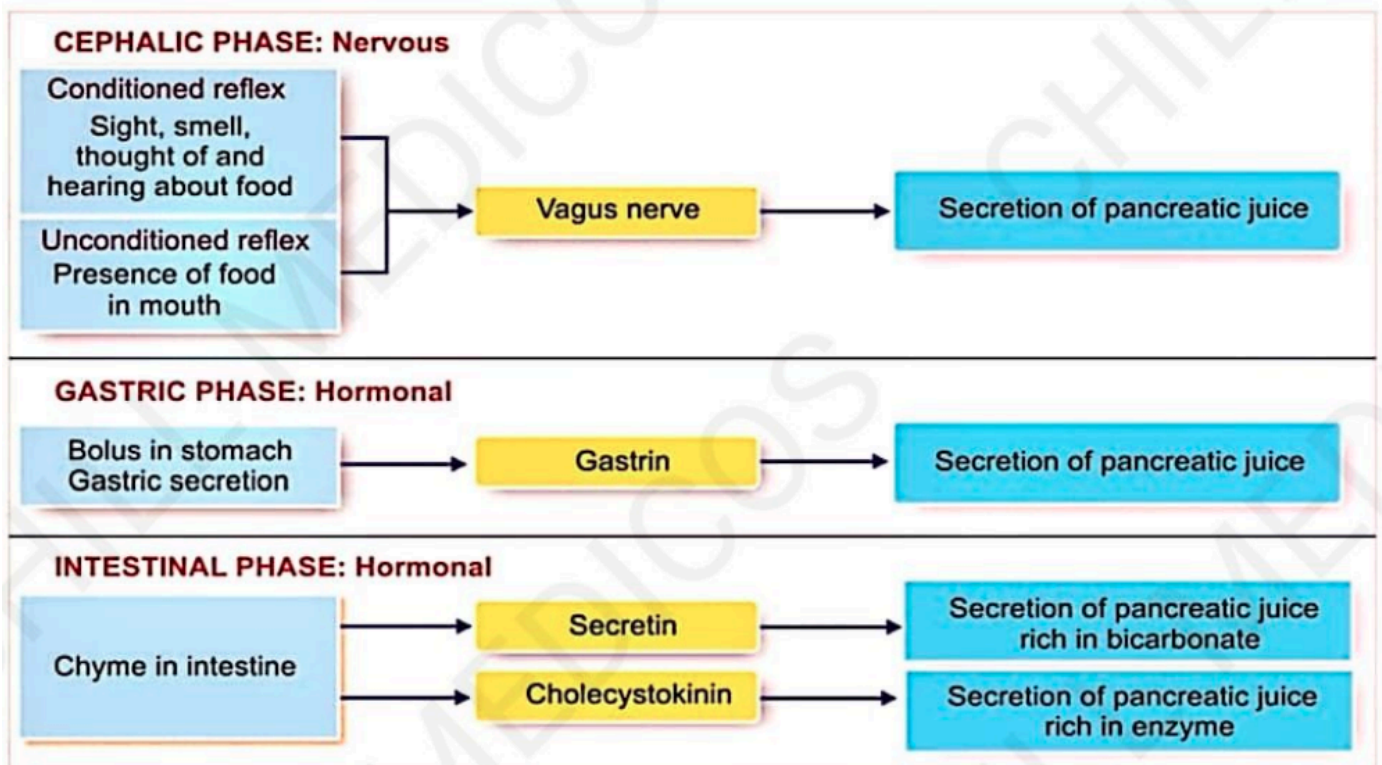


FIGURE 39.2: Schematic diagram showing the regulation of pancreatic secretion